

macOS Python Setup: pyenv + venv Recommended Workflow

1. Install pyenv

Install pyenv using Homebrew:

```
brew install pyenv
```

What it does

- Installs pyenv, a Python version manager

Why it's useful

- Lets you install multiple Python versions side by side
- Switch Python versions per project or globally
- Avoids breaking system Python (macOS relies on it)

2. Install a Python Version with pyenv

```
pyenv install 3.11.8
```

What it does

- Downloads, builds, and installs Python 3.11.8
- Installs it into: `~/.pyenv/versions/3.11.8/`

Important

- This does NOT change your current Python yet
- It just makes Python 3.11.8 available

List installed versions:

```
pyenv versions
```

3. Recommended Workflow (pyenv + venv)

Step 1: Set Python version for your project

```
pyenv local 3.11.8
```

Step 2: Create a virtual environment

```
python -m venv .venv
```

Step 3: Activate the virtual environment

```
source .venv/bin/activate
```

Step 4: Select interpreter in VS Code

Cmd + Shift + P → Python: Select Interpreter → Choose .venv/bin/python

Verification Commands

```
python --version  
which python  
pip list
```

Summary

- pyenv manages Python versions.
- venv isolates project packages.
- Together they provide safe, reproducible environments.

Quick Switching Summary

Goal (Detailed Explanation)	Command
Temporarily switch to system Python for current terminal only. Useful for quick tests without changing project settings.	<code>pyenv shell system</code>
Permanently switch back to system Python for all new terminals. Useful if you want system/Homebrew Python as default.	<code>pyenv global system</code>
Run system Python once without changing pyenv config. Useful for scripts or one-off commands.	<code>/usr/local/bin/python3</code>